Students table

Student_id	Name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

1. SELECT *
   FROM Students table
   Distinct department;

Department
IT
HR

2. SELECT department
   Avg(age) As Avg_age
   FROM Students table
   GROUP BY department;

Department	Avg_age
IT	20,5
HR	22
Finance	23

3 SELECT department
   FROM Students table
   GROUP BY department.
   HAVING COUNT >1;

Department	Students Count
IT	2
HR	2

4

SELECT student_id,
name,
age,
department
FROM Students table
WHERE age BETWEEN 21 AND 23;

Student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	finance
5	Eve	22	HR

5. SELECT student_id,
name,
age,
department
FROM Students ~~table~~
WHERE department IN ('IT'; 'HR')
AND age > 21;

Student_id	name	age	Department
2	Bob	22	HR
5	Eve	22	HR

6. SELECT department

2

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6. SELECT department
FROM Courses
GROUP BY department.
HAVING SUM(Credits) > 5 AS total credits

Department	total credits
IT	11

7 SELECT Course_id,
Course_name,
department,
Credits
FROM Course table
WHERE credits < > 4;

Course_id	Course_name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Powered by CamScanner

8

```
SELECT course_id,
       course_name,
       credits
FROM   Courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits	
102	Python	4	
103	Data Science.	4	
101	SQL Basics	3	

9. SELECT MAX,

enrollment table

enrollment_id	Student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

9

```
SELECT MAX(grade) AS max_grade,
       MIN (grade) AS min_grade,
       AVG (grade) AS avg_grade.
FROM   enrollmentstable;
```

Max grade	min grade	avg_grade
90	78	84,6.

10.
```
SELECT course_id,
       COUNT (*) AS enrollment count
FROM enrollment
GROUP BY course_id
```

course_id	enrollment count
101	1
102	1
103	1
104	1
105	1

Salaries table.

Employee_id	Name	dep	salary	
1	Tom	IT	60 000	5000.
2	Jerry	HR	55 000	4000
3	Spike	Finance	70 000	6 000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	35000

11 SELECT department
~~$~~ SUM(salary) AS total_salary
SUM(bonus) AS total_bonus
FROM salaries
GROUP BY department;

Department	total salary	total bonus
IT	122000	10500
HR	109 000	7500
finance	70 000	6000

12 SELECT department, AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING AVG (salary > 55 000;

Department	Avg_salary ~~Salaries~~
IT	61.000
finance	70 000

13

```
SELECT employee_id,
       name,
       salary,
       bons,
       (salary + bonus) AS total_compensation
FROM   salaries
WHERE  salary+bonus > 60 000
```

employee_id	name	salary	bonus	total compensa
1	Tom	60 000	5000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	62 000	5 500	67 500

14. Projects table.

Project_id	Project name	department	budget
1	AI App	IT	120 000
2	Payroll system	finance	80 000
3	Dashboard	IT	150 000
4	Website	Marketing	60 000
	HR Portal	HR	50 000

```
SELECT department,
       Sum(budget) AS total_budget,
       AVG (budget AS avg_budget
FROM   Projects
GROUP BY department
HAVING AVG (budget) > 70 000
```

department	total budget	avg budget
IT	270 000	135 000

15

```sql
SELECT Project_id,
       Project_name,
       department,
       budget
FROM   Projects
WHERE  budget BETWEEN 50000 AND 120000
  AND department <> 'Marketing';
```

Project_id	Project_name	department	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000